Refractive Index of a Prism

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1 Question 1

The difference between Fresnel and Fraunhofer diffraction is that the first one deals with when the observation plane is very close to the aperture, and deals with spherical wave fronts. The second kind deals with "far field" approximation. That is when the aperture to the plane distance is large. The far field approximation is satisfied because the collimator was providing parallel incident light. And also the telescope was focused at a far away object.

2 Question 2

Based on my observation, the pattern of the single slit was much more simple. It was one thicker line, a bit thicker, in the middle. Then the line's thickness and spacing were all even throughout. For the double slit it was quite weird, it was almost exactly like single slit except that there was a pattern of dimming in and out that had a horizontal frequency about 10 times lower than the frequency of the lines horizontally.

3 Question 3

When one is measuring many things are happening in this experiment that's not just the reading error. The scale reading is +/-0.5' but other things could introduce uncertainty for example how wide the opening slit is or what technique was used to measure the data. Crosshair positioned in bright or dark background could also affect the reading. Also if lab partners had different glasses prescription, one sharp image for one could be blurry for the other person.

Table 1: Single-slit diffraction minima. All angles are in degrees. $\theta_{L,R} = \phi_{L,R} - \phi_0$.

Trial	Order	Centre ϕ_0	Left ϕ_L	Right ϕ_R	θ_L	θ_R
1	1	84.867	85.000	84.750	0.133	-0.117
1	2	84.867	85.117	84.617	0.250	-0.250
1	3	84.867	85.317	84.500	0.450	-0.367
1	4	84.867	85.400	84.367	0.533	-0.500
1	5	84.867	85.550	84.217	0.683	-0.650
1	6	84.867	85.633	84.083	0.767	-0.783
2	1	84.867	85.000	84.683	0.133	-0.183
2	2	84.867	85.067	84.517	0.200	-0.350
2	3	84.867	85.200	84.433	0.333	-0.433
2	4	84.867	85.350	84.283	0.483	-0.583
2	5	84.867	85.500	84.167	0.633	-0.700

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Trial	Order	Centre ϕ_0	Left ϕ_L	Right ϕ_R	θ_L	θ_R
2	6	84.867	85.583	84.083	0.717	-0.783
3	1	84.850	85.000	84.700	0.150	-0.150
3	2	84.850	85.100	84.600	0.250	-0.250
3	3	84.850	85.250	84.500	0.400	-0.350
3	4	84.850	85.350	84.350	0.500	-0.500
3	5	84.850	85.500	84.200	0.650	-0.650
3	6	84.850	85.667	84.150	0.817	-0.700

Table 2: Double-slit diffraction minima. All angles are in degrees. $\theta_{L,R} = \phi_{L,R} - \phi_0$.

Trial	Order	Centre ϕ_0	Left ϕ_L	Right ϕ_R	θ_L	θ_R
1	1	84.900	85.250	84.433	0.350	-0.467
1	2	84.900	85.500	84.150	0.600	-0.750
1	3	84.900	85.933	83.733	1.033	-1.167
2	1	85.750	86.017	85.450	0.267	-0.300
2	2	85.750	86.500	85.000	0.750	-0.750
2	3	85.750	86.917	84.767	1.167	-0.983

Table 3: Double-slit interference maxima. All angles are in degrees. $\theta_{L,R} = \phi_{L,R} - \phi_0$.

Trial	Order	Centre ϕ_0	Left ϕ_L	Right ϕ_R	θ_L	θ_R
1	1	84.850	84.983	84.717	0.133	-0.133
1	2	84.850	85.017	84.583	0.167	-0.267
1	3	84.850	85.250	$84.933^\dagger$	0.400	$0.083^\dagger$
1	4	84.850	85.367	$84.783^\dagger$	0.517	$-0.067^\dagger$
1	5	84.850	85.517	84.150	0.667	-0.700
1	6	84.850	85.667	84.117	0.817	-0.733
2^*	1	84.850	84.983	—	0.133	—
2^*	2	84.850	85.017	—	0.167	—
2^*	3	84.850	85.250	—	0.400	—
2^*	4	84.850	85.367	—	0.517	—
2^*	5	84.850	85.517	—	0.667	—
2^*	6	84.850	85.667	—	0.817	—

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Trial	Order	Centre ϕ_0	Left ϕ_L	Right ϕ_R	θ_L	θ_R
3	1	85.467	85.550	85.283	0.083	-0.183
3	2	85.467	85.733	85.167	0.267	-0.300
3	3	85.467	85.850	85.000	0.383	-0.467
3	4	85.467	85.983	84.900	0.517	-0.567
3	5	85.467	86.150	84.767	0.683	-0.700
3	6	85.467	86.283	84.633	0.817	-0.833

† Right reading questionable: transcription or fringe order unverified.

* Incomplete trial; not used for paired left/right analysis.

Diffraction

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[2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from scipy.stats import chi2

#Import data
datafile = pd.read_excel('PHYS 2007 - Diffraction.xlsx',
                                    sheet_name = 'Single Slit Diffraction')
#Display data
display(datafile)
```

	Single Slit difffraction (Dark Spots)	Unnamed: 1	Unnamed: 2	Unnamed: 3
0	Trial 1	Left	NaN	Right
1	NaN	Degrees	Minutes	Degrees
2	Centre	84.5	22	NaN
3	1	85	0	84.5
4	2	85	7	84.5
5	3	85	19	84.5
6	4	85	24	84
7	5	85.5	3	84
8	6	85.5	8	84
9	NaN	NaN	NaN	NaN
10	Trial 2	Left	NaN	Right
11	NaN	Degrees	Minutes	Degrees
12	Centre	84.5	22	NaN
13	1	85	0	84.5
14	2	85	4	84.5
15	3	85	12	84
16	4	85	21	84
17	5	85	30	84
18	6	85.5	5	84
19	NaN	NaN	NaN	NaN
20	Trial 3	Left	NaN	Right
21	NaN	Degrees	Minutes	Degrees
22	Centre	84.5	21	NaN
23	1	85	0	84.5
24	2	85	6	84.5
25	3	85	15	84.5

```
26                                  4         85         21         84
27                                  5       85.5          0         84
28                                  6       85.5         10         84
```

```
   Unnamed: 4
0         NaN
1     Minutes
2         NaN
3          15
4           7
5           0
6          22
7          13
8           5
9         NaN
10        NaN
11    Minutes
12        NaN
13         11
14          1
15         26
16         17
17         10
18          5
19        NaN
20        NaN
21    Minutes
22        NaN
23         12
24          6
25          0
26         21
27         12
28          9
```

[16]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 1 Left
Ddata = datafile['Unnamed: 1'].iloc[3:8].to_numpy()
Mdata = datafile['Unnamed: 2'].iloc[3:8].to_numpy()

phi = np.mean(Ddata + Mdata/60)
#print(phi)

#Trial 1 Right
Ddata2 = datafile['Unnamed: 3'].iloc[3:8].to_numpy()
Mdata2 = datafile['Unnamed: 4'].iloc[3:8].to_numpy()
```

```
phi2 = np.mean(Ddata2 + Mdata2/60)

theta = (Ddata + Mdata/60)-(Ddata2 + Mdata2/60)

#print(phi2)
print(theta)
```

```
[0.25 0.5 0.8166666666666629 1.0333333333333456 1.3333333333333286]
```

[17]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 2 Left
Ddata3 = datafile['Unnamed: 1'].iloc[13:18].to_numpy()
Mdata3 = datafile['Unnamed: 2'].iloc[13:18].to_numpy()

phi3 = np.mean(Ddata3 + Mdata3/60)

#print(phi3)

#Trial 2 Right
Ddata4 = datafile['Unnamed: 3'].iloc[13:18].to_numpy()
Mdata4 = datafile['Unnamed: 4'].iloc[13:18].to_numpy()

phi4 = np.mean(Ddata4 + Mdata4/60)

theta2 = (Ddata3 + Mdata3/60)-(Ddata4 + Mdata4/60)

print(theta2)
```

```
85.22333333333333
[0.3166666666666629 0.5499999999999972 0.7666666666666657
 1.0666666666666629 1.3333333333333286]
```

[14]:
```
#Grabing numbers code and summing the angle readings
#Single Slit Diffraction (Dark Spots)
#Trial 3 Left
Ddata5 = datafile['Unnamed: 1'].iloc[13:18].to_numpy()
Mdata5 = datafile['Unnamed: 2'].iloc[13:18].to_numpy()

phi5 = np.mean(Ddata5 + Mdata5/60)
print(phi5)

#Trial 3 Right
Ddata6 = datafile['Unnamed: 3'].iloc[13:18].to_numpy()
Mdata6 = datafile['Unnamed: 4'].iloc[13:18].to_numpy()

phi6 = np.mean(Ddata6 + Mdata6/60)
```

```
print(phi6)
```

```
85.22333333333333
84.41666666666666
```

[ ]: